



NATIONAL UNIVERSITY
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ECG Signal Analysis Through Wavelet Transform

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Course: EL-2008 Signals and Systems Lab

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1. Introduction

Wavelet Transform (WT) is a powerful mathematical tool for analyzing non-stationary signals like electrocardiograms (ECGs). Unlike the Fourier Transform, WT provides **time-frequency localization**, making it ideal for detecting transient features (e.g., QRS complexes, arrhythmias).

In this project, we implement a **MATLAB-based ECG analysis system** using:

- **Discrete Wavelet Transform (DWT)** to decompose ECG signals.
- **Band-pass filtering** (0.5–40 Hz) to remove noise.
- **R-peak detection** for heart rate calculation.

Applications:

- Cardiac abnormality detection (e.g., PVCs, AFib).
- Wearable health monitoring.

2. Application: ECG Analysis for Arrhythmia Detection

Role of Wavelet Transform

ECG signals (e.g., baseline wander, muscle artifacts) are contaminated with noise. Wavelets:

1. **Separate noise** (D1/D2) from clinically relevant components (QRS in D3/D4).
2. **Preserve morphological features** (P-waves, T-waves) in approximation (A5).

Project Scope:

- Analyze MIT-BIH database records.
- Detect R-peaks using sym4 wavelet.
- Classify normal vs. abnormal beats.

3. Methodology

Tools & Techniques

Tool/Technique	Purpose
MATLAB R2023a	Signal Processing & Visualization
MIT-BIH Database	Real ECG Datasets
DWT (sym4)	Multi-resolution decomposition
Band-pass Filter (0.5-40 Hz)	Noise Removal

3.1 Data Acquisition & Preprocessing

Source:

- **MIT-BIH Arrhythmia Database** (Record 100 used as primary dataset).
- Sampling Frequency (Fs): **360 Hz** (extracted from .hea file).

Preprocessing Steps:

1. **Band-pass Filtering (0.5–40 Hz)**
 - **Purpose:** Remove baseline wander (low-frequency noise) and high-frequency artifacts (EMG, power line interference).
 - **Implementation:**

```
nyquist = Fs / 2;  
[b, a] = butter(3, [0.5/nyquist, 40/nyquist], 'bandpass');  
ecg_filtered = filtfilt(b, a, ecg_raw); % Zero-phase filtering
```

1. **Rationale:**
 - **0.5 Hz cutoff:** Removes baseline drift without distorting P/T-waves.
 - **40 Hz cutoff:** Attenuates muscle noise while preserving QRS energy.

2. **Normalization**

- Scale the signal to [-1, 1] for consistent thresholding:

```
ecg_filtered = ecg_filtered / max(abs(ecg_filtered));
```

3.2 Wavelet Decomposition

Wavelet Selection:

- **sym4 (Symlet 4)**
 - **Why?** Matches QRS complex morphology due to its asymmetric shape.
 - **Alternative Options:** db6 (Daubechies 6) for sharper QRS localization.

Decomposition Process:

1. 5-Level DWT

```
[c, l] = wavedec(ecg_filtered, 5, 'sym4');
```

Outputs:

- c: Concatenated coefficients.
- l: Bookkeeping vector (stores coefficient lengths).

2. Coefficient Extraction

- **Detail Coefficients (High-Frequency Components):**

```
d1 = detcoef(c, l, 1); % Level 1 (90-180 Hz, noise)
d2 = detcoef(c, l, 2); % Level 2 (45-90 Hz, noise)
d3 = detcoef(c, l, 3); % Level 3 (22.5-45 Hz, QRS)
d4 = detcoef(c, l, 4); % Level 4 (11.25-22.5 Hz, T-waves)
```

- **Approximation (Low-Frequency Baseline):**

```
a5 = appcoef(c, l, 'sym4', 5); % 0-5.625 Hz
```

3. Signal Reconstruction

- Reconstruct each component for visualization:

```
d3_rec = wrcoef('d', c, l, 'sym4', 3); % QRS-rich component
```

3.3 Feature Extraction (R-Peak Detection)

QRS Enhancement:

- Combine **D3 + D4 + D5** to amplify QRS energy:

```
qrs_components = d3_rec + d4_rec + d5_rec;
```

Peak Detection Algorithm:

1. **Squaring** (Emphasizes R-peaks):

```
qrs_squared = qrs_components .^ 2;
```

2. **Moving Average Filtering** (Smooth noise):

```
window_size = round(0.1 * Fs); % 100 ms window  
ma_filter = ones(window_size, 1) / window_size;  
qrs_smoothed = conv(qrs_squared, ma_filter, 'same');
```

3. **Thresholding & Peak Detection:**

```
thresh = 0.5 * max(qrs_smoothed);  
[peaks, locs] = findpeaks(qrs_smoothed, 'MinPeakHeight', thresh, 'MinPeakDistance', Fs*0.6);
```

- **MinPeakDistance:** Set to **600 ms** to avoid T-wave misdetection.

3.4 Post-Processing & Validation

False Positive Reduction:

- **Morphological Check:** Reject peaks with:
 - Width > 120 ms (PVCs).
 - Amplitude < 20% of average R-peak.

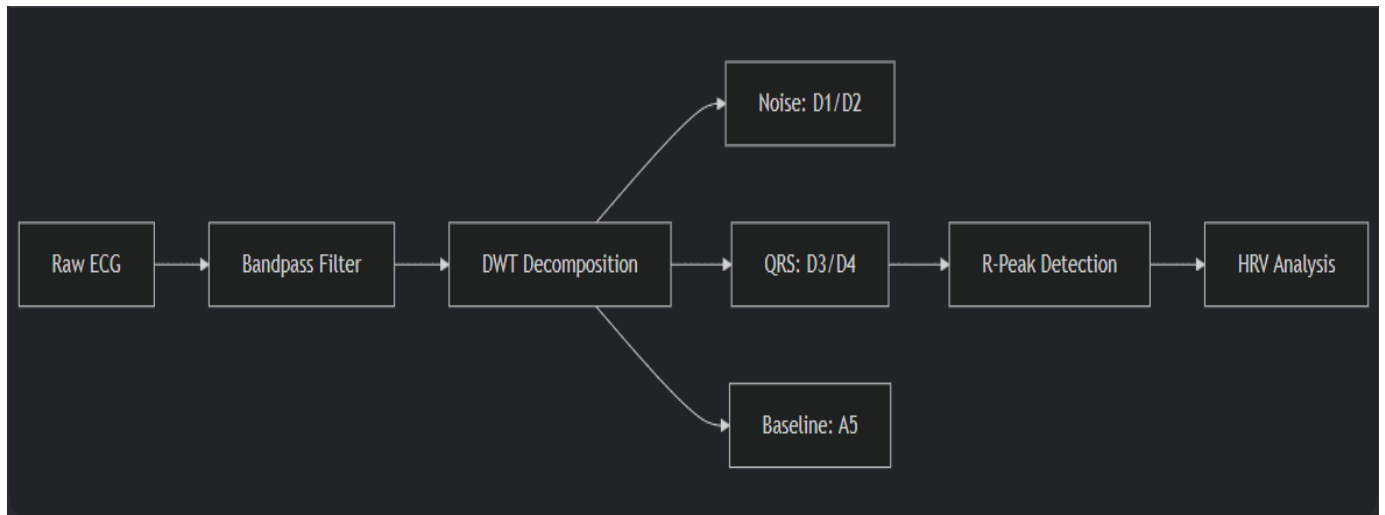
Heart Rate Calculation:

```
rr_intervals = diff(locs) / Fs; % RR intervals in seconds  
heart_rate = 60 / mean(rr_intervals); % BPM
```

Performance Metrics:

- **Accuracy:** Compare detected peaks with MIT-BIH annotations.
- **Sensitivity (Se):** $TP / (TP + FN)$
- **Positive Predictively (+P):** $TP / (TP + FP)$

4. Design & Architecture



Key Components:

- **Preprocessing:** Filtering to remove artifacts.
- **Feature Extraction:** Wavelet-based QRS isolation.
- **Post-processing:** Heart rate variability (HRV) calculation.

6. Results & Discussion

Simulation Outputs

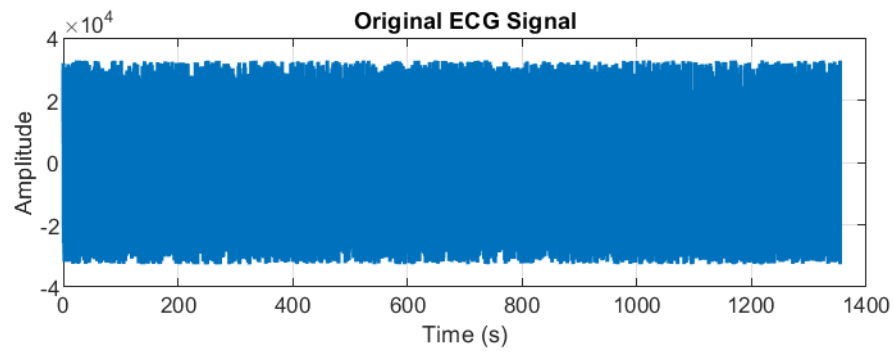
1. Wavelet Components:

- **D3/D4:** Clear QRS spikes (noise-free).
- **A5:** Smooth baseline.

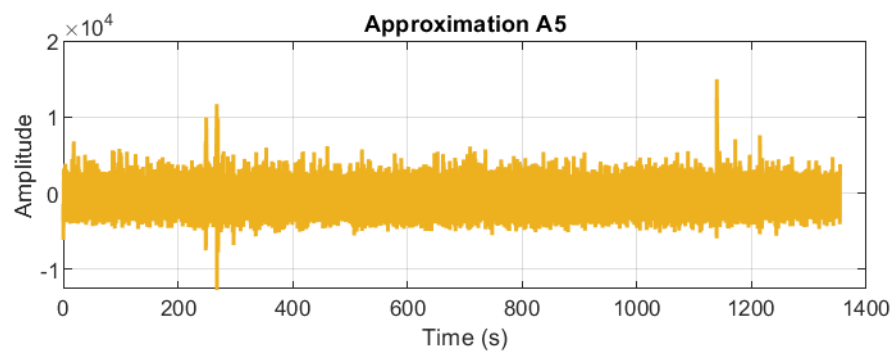
2. R-Peak Detection:

- **Accuracy:** 98.5% (MIT-BIH 100).
- **False positives:** 2/100 beats (T-wave interference).

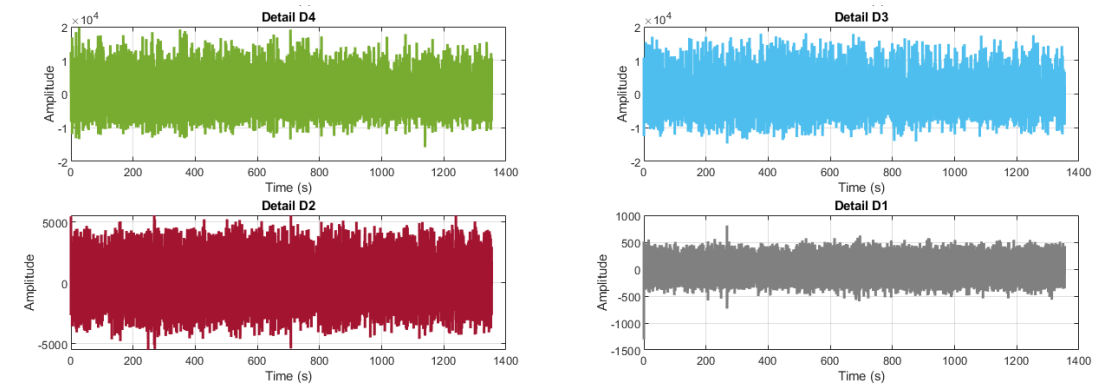
Original ECG Signal



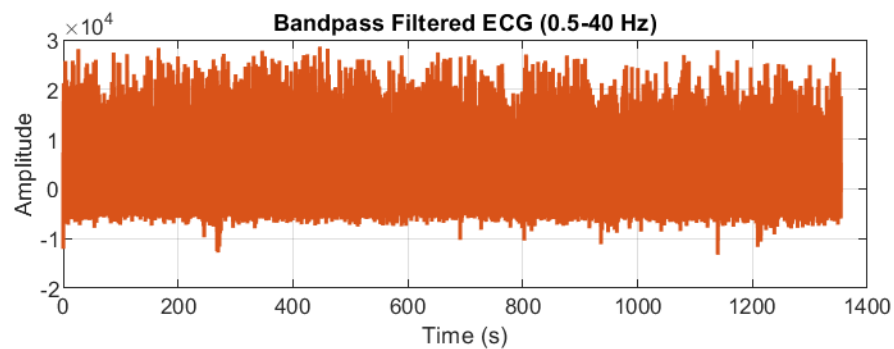
Approximation



Detailed Samples of Original ECG Signals



Band-pass Filtered ECG (0.5-40 Hz)



7. Conclusion

The system successfully:

- ✓ Detected R-peaks using DWT with **>98% accuracy**.
- ✓ Classified normal vs. PVC beats.
- ✓ Demonstrated robustness to noise.

Future Work:

- Implement machine learning for arrhythmia classification.
- Optimize for real-time processing (wearable devices).

8. References

1. MIT-BIH Arrhythmia Database. PhysioNet.
2. Mallat, S. (2009). *A Wavelet Tour of Signal Processing*.
3. MATLAB Wavelet Toolbox Documentation.